

CSE-3523: Microprocessors, Microcontrollers & Embedded Systems

Midterm Syllabus Summary (Autumn-25)

Segment 1

Focus Area:

- Introduction to Microprocessor and its main tasks
- Microcontroller vs microprocessor
- Generation of Microprocessor and its types
- Hardware Components of a Microprocessor
- System Bus (Data Bus, Address Bus, Control Bus – Overview)
- Instruction Execution Cycle
- Memory Addressing Techniques

Segment 2

This segment includes the **entire content** of Segment 2 from the Lecture material.

Segment 3 (Part-1)

Focus Area:

- Assembly Language
- Assembly Language syntax :Rules,Program Statement,Program data(byte variable ,word variable) program structure,memory model,procedure

Segment 3 (Part-2)

Focus Area:

- Data Transfer Instructions: MOV, PUSH, POP
- Arithmetic Instructions: ADD, INC, ADC, MUL
- Logical Instructions: AND, OR, XOR,SHIFT

*focus on understanding the concepts rather than rote memorization. Practice writing programs and solving problems involving **arithmetic, logical, and data transfer instructions, as well as addressing modes, registers, and flags**. Clear understanding and regular practice will help you answer questions accurately and score higher marks.*